



DATABASE

**InTxDB: interaction data between gram-negative bacteria
secreted effectors and host proteins**

Journal:	<i>DATABASE</i>
Manuscript ID	DATABASE-2025-0034
Manuscript Type:	Original Article
Date Submitted by the Author:	18-Mar-2025
Complete List of Authors:	Zhu, Yanyan; Zhejiang University College of Life Sciences, Department of Bioinformatics Liu, Liya; Zhejiang University, College of Life Science Hu, Yueming; Zhejiang University College of Life Sciences, Department of Bioinformatics Li, Sida; Zhejiang University College of Life Sciences, Department of Bioinformatics Liu, Enyan; Zhejiang University College of Life Sciences, Department of Bioinformatics Hu, Yanshi; Zhejiang University - Zijingang Campus, Department of Bioinformatics; State Key Laboratory of Plant Physiology and Biochemistry, College of Life Sciences; Zhang, Shilong; Zhejiang University College of Life Sciences, Department of Bioinformatics Chao, Haoyu; Zhejiang University, Department of Bioinformatics Fang, Qiuyu; Zhejiang University College of Life Sciences, Department of Bioinformatics Yu, Huan; Zhejiang University College of Life Sciences, Department of Bioinformatics Chen, Ming; Zhejiang University, Bioinformatics
Keywords:	protein-protein interaction, protein docking, Bacterial secretion system effector, InTxDB

SCHOLARONE™
Manuscripts

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

**InTxDB: interaction data between gram-negative bacteria
secreted effectors and host proteins**

Yanyan Zhu^{1, #}, Liya Liu^{1, #}, Yueming Hu^{1, #}, Sida Li¹, Enyan Liu¹, Yanshi Hu¹, Shilong Zhang¹, Haoyu Chao¹, Qiuyu Fang², Yu Huan², Ming Chen^{1, *}

¹ Department of Bioinformatics, College of Life Sciences, Zhejiang University, Hangzhou, China.

² Department of Thoracic Surgery, The Second Affiliated Hospital Zhejiang University School of Medicine, Zhejiang University, Hangzhou, Zhejiang Province, People's Republic of China.

Y.Z., L.L. and Y.H. contributed equally to this study.

*Correspondence: mchen@zju.edu.cn (M.C.); Tel./Fax: +86-(0)571-88206612 (M.C.)

Abstract

Gram-negative bacteria utilize specialized secretion systems to deliver effectors into the cytoplasm of eukaryotic cells, where they manipulate host cellular functions to promote infection. These interactions play a crucial role in bacterial pathogenesis and poses significant public health challenges. Understanding of the protein-protein interactions (PPIs) between effectors and host proteins is essential for deciphering infection mechanisms and developing novel therapeutic strategies. However, existing databases on pathogen-host interactions often lack comprehensive coverage of secreted effectors and their host targets, limiting their utility in studying bacterial infections. Furthermore, current resources are often incomplete, poorly annotated, and lack predictive capabilities. To address these limitations, we have developed InTxDB, a comprehensive database dedicated to interactions involving bacterial type I-X secreted effectors and their host counterparts. InTxDB currently includes 1,853 experimentally validated interaction pairs from 102 bacterial species, with detailed annotations on protein sequences, functions, subcellular localizations, secretion signals, structures, and interaction sites. By filling a critical gap in existing databases, InTxDB provides a well-curated and functionally enriched resource to advance research on bacterial pathogenesis and host-pathogen interactions.

Keywords: protein-protein interaction; protein docking; Bacterial secretion system effector; InTxDB

Introduction

Gram-negative bacteria utilize specialized secretion systems to transport proteins across their cell membranes, with at least six distinct systems identified [1]. These secretion systems can be classified into two main types. The first type, such as the Type III secretion system (T3SS), enables the direct translocation of substrate proteins into eukaryotic host cells in a single step [2]. The second type, including the Sec and Tat pathways, initially transports substrate proteins to the periplasm, followed by a secondary mechanism that transfer across the outer membrane or into the host cell [3]. The translocated substrate proteins, known as effectors, play a critical role in bacterial infection and pathogenicity by manipulating host cellular processes to promote survival and proliferation [1],[3].

Once translocated into the host cytoplasm, bacterial effectors disrupt essential signaling pathways, leading to disease. For example, the T3SS effector YopJ, secreted by *Yersinia pestis*, inhibits the MAPK and NF- κ B signaling pathways, suppressing the host inflammatory response and allowing immune detection [4]. The T4SS effector CagA, produced by *Helicobacter pylori*, binds to the SH2 domain of host proteins, activating SHP-2 phosphatase and inducing sustained ERK/MAPK signaling pathway, which promotes abnormal gastric epithelial cell proliferation and contributes to gastric cancer development [5]. Additionally, the T6SS effector VgrG, secreted by *Pseudomonas aeruginosa*, promotes inflammasome assembly and activates the NLRP3 signaling pathway, triggering the release of pro-inflammatory cytokines such as IL-1 β , leading to a strong inflammatory response [6].

Over the past few decades, experimental techniques such as co-immunoprecipitation (Co-IP) and high-throughput methods like mass spectrometry (MS) and yeast two-hybrid (Y2H) [7]-[9] have significantly expanded our understanding of protein-protein interactions (PPIs) between gram-negative bacterial effectors and host proteins. These techniques curated in several host-pathogen protein interaction databases[10]-[12]. For example, PHISTO integrates host-pathogen PPIs and provides functions annotations of pathogen-targeted human proteins [10]. STRING offers extensive PPIs datasets with confidence score to assess interaction reliability [11]. IntAct provides detailed biological context and functional

1
2
3
4 annotations for each interaction [12]. However, none of these databases specifically focus
5
6 on interactions between bacterial effectors, and they often lack atomic-level details, such
7
8 as binding sites and structural information.
9

10 To address this gap, we have developed InTxDB (<http://bis.zju.edu.cn/InTxDB>), a
11
12 dedicated database for gram-negative bacteria type I-X secreted effectors and their host
13
14 interactions. InTxDB currently contains 1,853 experimentally validated interaction pairs
15
16 from 102 bacterial species, with comprehensive annotations on protein sequences,
17
18 functions, subcellular localizations, secretion signals, structures, and interaction sites. By
19
20 providing a well-curated and functionally enriched dataset, InTxDB serves as a valuable
21
22 resource for studying bacterial pathogenesis and host-pathogen interactions, facilitating
23
24 further research and therapeutic development.
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Result

Overall description of InTxDB

InTxDB is a comprehensive database dedicated to interactions between gram-negative bacterial type I-X secreted effectors (TxSEs) and their human host proteins. It currently contains 1,853 experimentally validated interaction pairs from 102 bacterial species (Table 1). Each entry is extensively annotated with information on protein sequences, functions, subcellular localizations, secretion signals, structural data, and interaction sites.

Table 1. The number of PPIs and the number of bacteria effectors, bacterial species, and human proteins of each interaction.

Type	Number of bacterial effectors	Number of human protein	Number of bacterial species	Number of PPIs
T1SE	11	24	16	40
T2SE	4	5	3	6
T3SE	58	767	16	1228
T4SE	58	276	24	414
T5SE	30	37	28	62
T6SE	3	10	1	10
T7SE	17	55	13	62
T8SE	4	2	3	5
T9SE	5	6	3	8
T10SE	3	11	3	18

The database architecture consists of three main components: data collection, data processing, and web construction (Figure 1). Data collection involves extracting experimentally validated PPIs from literature and databases such as IntAct[12], PHISTO[10], and HPIDB3.0[30], while protein structures are sourced from RCSB PDB[33] and AlphaFold DB[34]. Data processing includes transmembrane helix and signal peptide prediction (TMHMM2.0[37], SignalP6.0[38]), protein docking simulations (ZDOCK[13], HDOCK[14], AlphaFold3[15]), and interaction site analysis (PDBePISA[16]). Finally, in the web construction phase, users can explore interaction data through a searchable and interactive interface, with tools such as Diamond[17] and HPInet (Hu, et al., unpublished), alongside statistical insights into bacterial species, taxonomy, and interaction distributions.

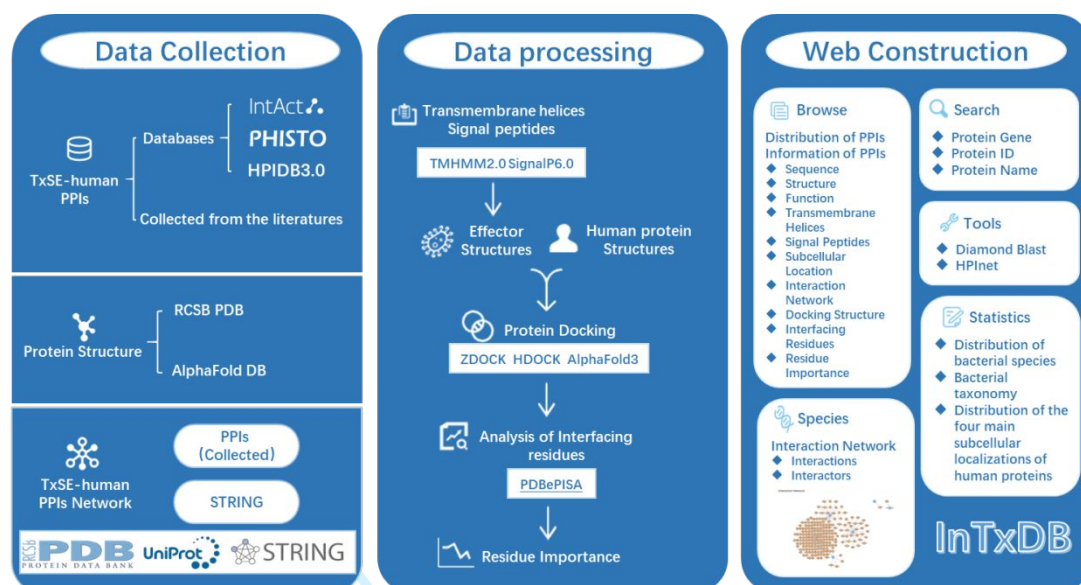


Figure 1. The workflow of InTxDB. The construction of InTxDB involves three main stages: **data collection, data processing, and web construction.** Experimentally validated effector-host PPIs were integrated from multiple databases, protein structures were obtained from structural repositories, and computational tools were applied for interaction prediction and analysis. The web platform provides interactive browsing, search functionalities, and analysis tools, offering a comprehensive resource for studying bacterial effector-host interactions.

By integrating high-confidence experimental data, structural predictions, and computational tools, InTxDB serves as a valuable resource for studying bacterial pathogenesis and host-pathogen interactions.

User Interface of InTxDB

InTxDB provides a user-friendly interface for exploring effector proteins and their interactions through multiple search options. Figure 2 illustrates the Home Page, which features a quick search bar, allowing users to retrieve proteins using gene names, protein names, or protein IDs. Additionally, the homepage presents statistical summaries of effectors and their interactions, along with a visual representation of the interaction network, offering an intuitive overview of the database's content and connectivity.

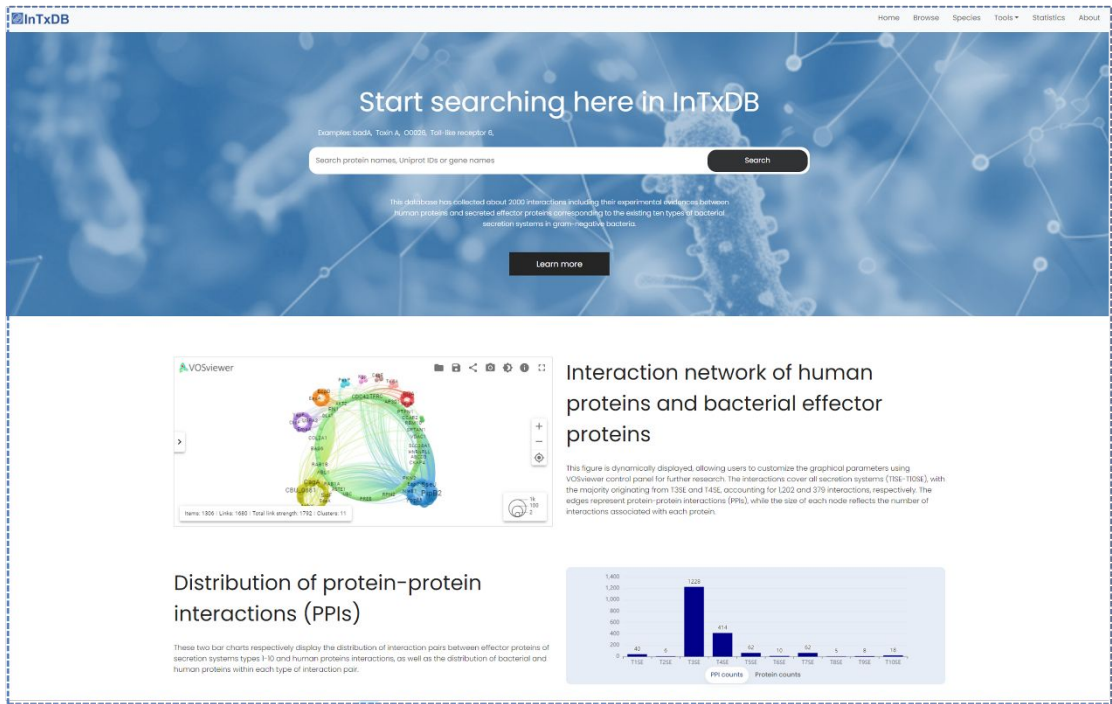


Figure 2. InTxDB database home page and statistics of datasets.

Browse and Data Exploration

InTxDB offers a user-friendly with an intuitive filtering system. Users can refine search results using filters on the left sidebar, including bacterial species, secretion system type, and protein length, enabling quick access to relevant information (Figure 3A).

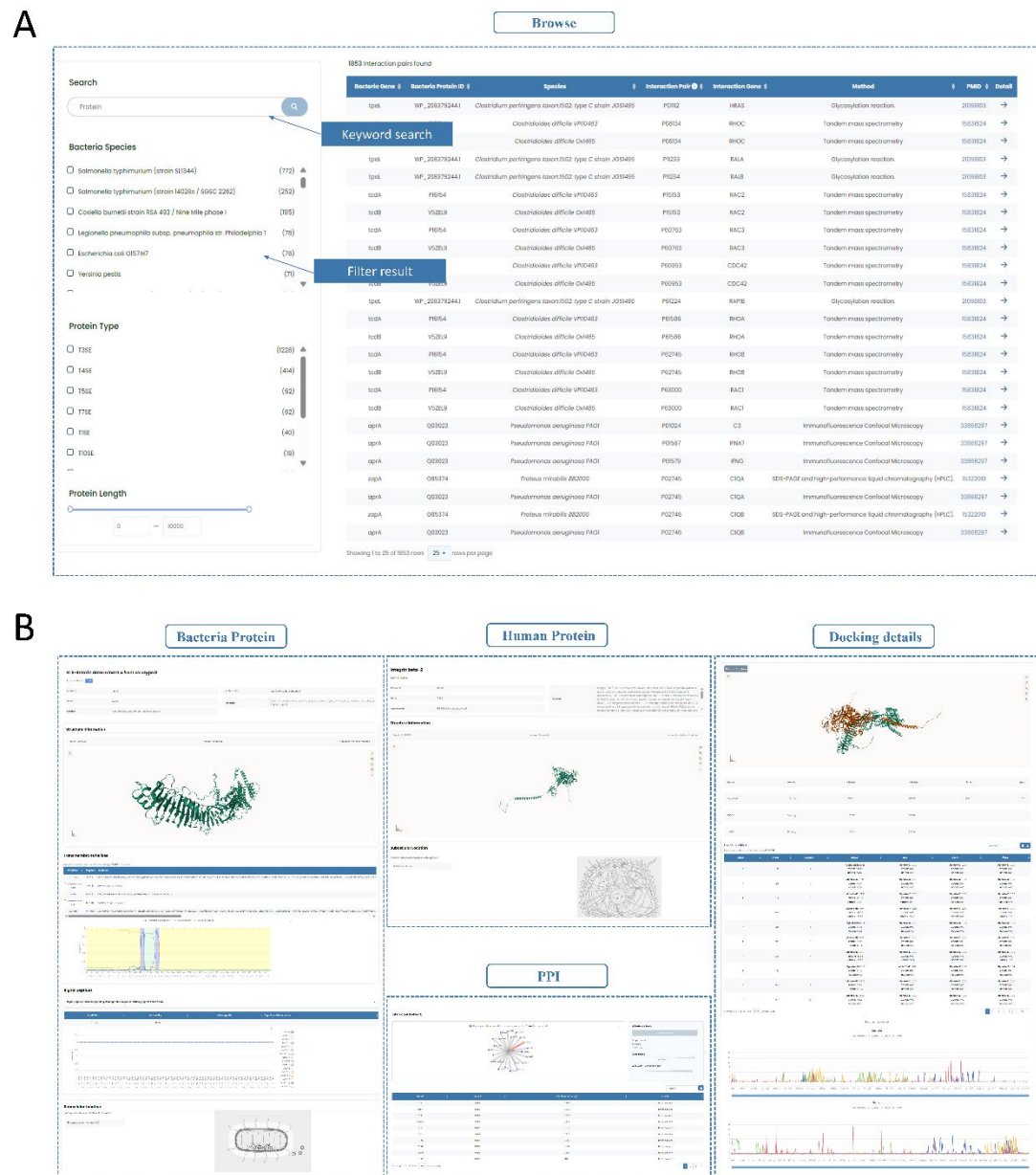
Upon selecting an entry, users are directed to a detailed information page, organized into four main sections: Bacterial Protein, Human Protein, PPI (Protein-Protein Interaction), and Docking Details (Figure 3B). These sections can be easily accessed via the sidebar navigation:

Bacterial Protein & Human Protein Sections: Display fundamental details, including protein structures. The Bacterial Protein section provides additional annotations such as signal peptides and transmembrane helices.

PPI Section: Presents the interaction network, offering insights into the relationships between effectors and human proteins, along with detailed information for each node.

Docking Details section: Showcases predicted docking generated by ZDOCK [13], HDOCK [14], and AlphaFold3 [15], along with interaction site analysis from PDBePISA [16]

and an assessment of residue importance for both interacting proteins.



Species-Specific Interaction Networks

InTxDB offers species-specific interaction networks, allowing users to explore bacterial effector-host interactions with enhanced filtering options. Users can refine results using the filter bar in the top left, selecting criteria such as species, interaction detection method, and

effector type. Filtered entries can be toggled between "Interactions" and "Interactors", providing flexibility in data visualization. Each interaction or protein entry links to a detailed information page, enabling deeper insights into bacterial pathogenesis (Figure 4).



Figure 4. Species of InTxDB. InTxDB provides species-specific interaction networks, allowing users to explore bacterial effector-host interactions across different bacterial species. Users can filter results by species, interaction detection method, and effector type, with options to toggle between "Interactions" and "Interactors" for a more detailed analysis of the dataset.

Tool

InTxDB provides two integrated data analysis tools to facilitate interaction prediction and sequence alignment. The Diamond [17] tool enables users to perform sequence alignment with customizable parameters and provides downloadable results (Figure 5A). The HPInet

Prediction tool (Hu, *et al.*, unpublished) allows users to analyze interactions based on bacterial protein sequences, human protein sequences, or predefined PPI lists. The tool generates network structure visualizations and residue importance heatmaps, offering insights into key interaction sites (Figure 5B). These tools enhance the functionality of InTxDB, supporting computational predictions and deeper analysis of bacterial effector-host interactions.

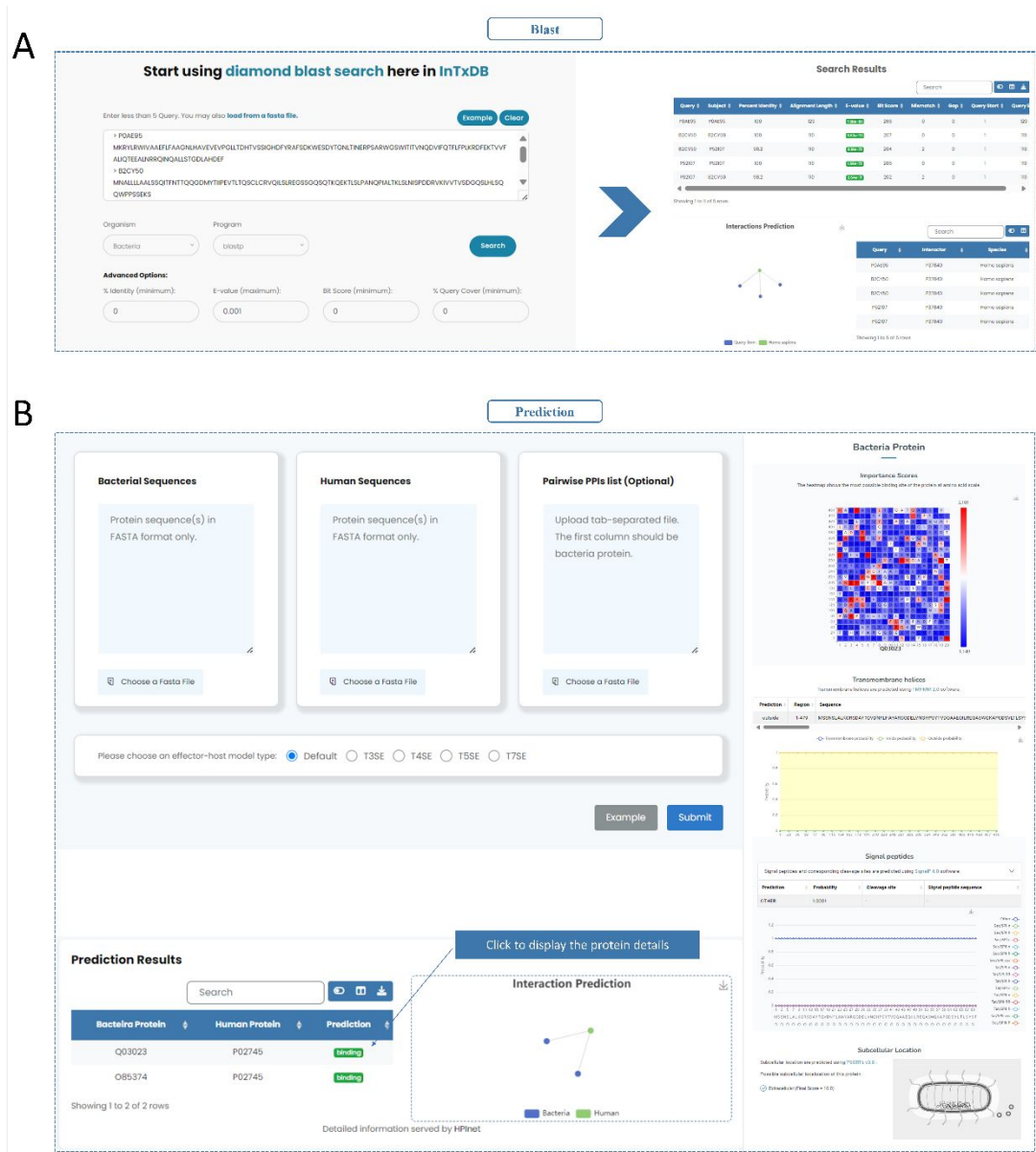


Figure 5. Tools of InTxDB. (A) Diamond Tool: Provides sequence alignment capabilities with customizable parameters, allowing users to analyze and download alignment results. (B) HPInet Prediction Tool: Predicts bacterial effector-host interactions based on sequence data or predefined PPI lists, generating network visualizations and residue importance heatmaps for deeper interaction analysis.

Statistical Insights

The Statistics section displays the distribution of bacterial species whose secretion effectors interact with host proteins, the classification of bacterial proteins, and the distribution of the four main subcellular localizations of human proteins that interact with effector proteins from different bacterial secretion systems.

Application in biological research

InTxDB serves as a valuable resource for biological research, such as investigating how LtxA, a leukotoxin secreted by *Aggregatibacter actinomycetemcomitans*, contributes to disease. By searching LtxA in InTxDB, users can identify its interaction with Integrin beta-2, a key immune cell receptor. Through PPI network analysis, it can be inferred that LtxA disrupts immune cell function, supporting previous findings [18]. Additionally, docking results help predict potential interaction targets, which may aid in developing novel biologic therapies for conditions such as leukemia, AIDS, and autoimmune disorders [19]-[21].

Discussion

InTxDB provides a comprehensive resource for understanding the complex interactions between bacterial secretion effectors and host proteins, which are central to the pathogenesis of many bacterial infections. Unlike existing databases like IntAct [12] and PHISTO [10], which cover general host-pathogen interactions, InTxDB specifically focuses on bacterial secretion system effectors and their human targets. Additionally, it offers three-dimensional protein structures, docking models, and interaction site analyses, facilitating a deeper understanding of microbial virulence mechanisms and the identification of potential therapeutic targets.

Recent studies have emphasized the significance of bacterial effector proteins in manipulating host cellular processes to promote infection [22]-[25]. Among these, the type III secretion system (T3SS) effectors from *Salmonella spp.* play a pivotal role in hijacking host cell machinery to facilitate bacterial survival and replication [23]-[25]. For instance, the T3SS effector SopE2 reprograms macrophage metabolism, enabling *Salmonella Typhimurium* to enhance intracellular replication and virulence [23]. Similar interactions have been studied for many other bacterial pathogens, demonstrating the widespread role of T3SS effectors in manipulating host cellular functions to promote infection [24]. InTxDB offers detailed data on these interactions, enabling researchers to explore not only the effectors involved but also their molecular mechanisms. The classification of bacterial proteins and their corresponding effector types can help researchers identify potential virulence factors and therapeutic targets [26],[27].

The HPI-net prediction tool in InTxDB enhances the ability to predict protein-protein interactions (PPI) by leveraging known sequence data (Hu, *et al.*, unpublished). By integrating experimental data with predictive modeling, InTxDB enables users to make informed hypotheses about the roles of uncharacterized effectors or host proteins in bacterial infections.

Moreover, InTxDB provides subcellular localization data for human proteins interacting with bacterial effectors, which is crucial for understanding how these interactions influence cellular processes. Recent studies have demonstrated that the localization of host proteins

plays a key role in bacterial infection strategies [28],[29]. For instance, the nuclear localization of the T4SS effector protein CBU0781 (AnkG) from *Coxiella burnetii* is essential for inhibiting apoptosis [29]. The anti-apoptotic activity of AnkG is regulated by p32-mediated intracellular transport, highlighting the importance of p32 localization in suppressing apoptosis during bacterial infection [29]. The subcellular localization statistics provided by InTxDB contribute to this growing body of knowledge. Understanding where these interactions occur within host cells is critical for designing therapeutic strategies that can block bacterial entry or disrupt bacterial manipulation of host cellular functions.

In conclusion, InTxDB serves as an essential resource for researchers in microbiology, immunology, and drug discovery. By providing access to high-quality datasets on bacterial effector-host protein interactions, along with powerful sequence alignment and predictive modeling tools, InTxDB accelerates research into bacterial pathogenesis and supports the discovery of novel therapeutic targets for combating bacterial infections.

Conclusions

InTxDB provides a comprehensive and user-friendly platform for exploring the interactions between bacterial secretion effectors and host proteins, facilitating a deeper understanding of host-pathogen interactions. The database offers a robust suite of tools, including sequence alignment and prediction features such as Diamond[17] and HPInet (Hu, *et al.*, unpublished). These tools empower researchers to easily access, query, and analyze complex interaction data, uncovering valuable insights into the molecular mechanisms underlying bacterial pathogenesis. The inclusion of key statistics on bacterial species distribution, protein classification, and subcellular localization of human proteins further enriches the database, offering a holistic view of bacterial effector-host protein interactions. InTxDB is a valuable resource for the scientific community, enabling both basic and applied research in microbial pathogenesis, drug discovery, and therapeutic intervention strategies.

Materials and methods

Collection of experimentally verified effector-human PPI data

We collected experimentally validated protein-protein interactions (PPIs) from three public databases: HPIDB [30], PHISTO [10], and IntAct [12], as well as recently published literature. First, we obtained experimentally validated effectors from two public databases: SecReT4 [31] and TxSEdb (<http://61.160.194.165:3080/TxSEdb/>) [32]. Next, the gene names of these effectors and their corresponding bacterial species were used as search terms to retrieve relevant PPI-related literature. Candidate articles from the search results were manually screened and curated, leading to the final collection of experimentally verified PPIs.

Three-dimensional complex structures of effector-human PPIs

The three-dimensional structures of bacterial effectors and human proteins were sourced from the PDB [33] databases and AlphaFold [34] databases. Selection criteria mandated that the protein structures must contain full-length sequences. In cases where the required structures were unavailable in the PDB databases, they were obtained from the AlphaFold database.

The three-dimensional structures of effector-human PPIs were generated through docking simulations using three software tools: ZDOCK [13], HDOCK [14], and AlphaFold3 [15]. ZDOCK and HDOCK performed docking using Protein Data Bank files, whereas AlphaFold3 utilized protein sequences. Protein sequences were retrieved from UniProt [35] and NCBI [36] databases. For proteins with sequences exceeding 1500 amino acids in length, only the first 1500 amino acids were selected for docking.

Prediction of effector transmembrane helices, signaling peptides, and subcellular localization

Transmembrane helices were predicted using TMHMM 2.0 [37] software, and signal peptides along with their cleavage sites were predicted using SignalP 6.0 [38] software. Subcellular localization data were retrieved from the UniProt database. For proteins with

unknown subcellular localization, predictions were generated using PSORTb v3.0 [39].

Interaction interface analysis

Interfacing residues are predicted using PDBePISA [16]. Residue importance was analyzed using PDBePISA, and the BURIEDSURFACEAREAScore values were normalized by first subtracting the mean from each value and then dividing by the standard deviation.

Network construction

STRING (<https://string-db.org/>) [11] is a database that systematically collects and integrates protein-protein interactions, including physical interactions and functional associations. All interaction networks associated with the specified protein were downloaded from the STRING database by selecting the protein species and inputting the protein sequence or protein name. User can freely set the confidence level to select the desired interaction network. The resulting interaction dataset was exported in a compatible TSV format for further organization and network integration.

Database construction

InTxDB utilizes Django as the backend web framework. A MySQL service is employed to store data tables, including the basic information of bacterial effectors and host protein-protein interaction (PPI) entries. Network visualization is interactively presented using Cytoscape.js [40] and ECharts.js [41] plugins. The Mol* (/molstar/) [42], a modern web-based open-source tool, is used for the visualization and analysis of large-scale molecular data, particularly for displaying three-dimensional protein structures.

Data availability

Availability

The online version of HPInet is freely accessible at <http://bis.zju.edu.cn/InTxDB>.

Funding

This work was supported by the National Key Research and Development Program of China [2023YFE0112300]; National Natural Sciences Foundation of China [32070677, 32270709, 32261133526]; 151 Talent Project, and Science and Technology Innovation Leader of Zhejiang Province [2022R52035]; Jiangsu Collaborative Innovation Center for Modern Crop Production and Collaborative Innovation Center for Modern Crop Production co-sponsored by province and ministry.

Authors' Contribution

MC, and YH conceived and supervised the project. YZ, YH, LL, and MC coordinated the project. YZ dataset collection. YZ, YH, and LL provided codes, models and software tools. YZ, and LL developed the website. YZ, YH, LL, EL, SL, YSH, SZ, HC, QF, YH, and MC wrote the first draft of this manuscript. YZ, YH, LL, and MC revised the manuscript accordingly.

Conflict of Interest: none declared.

References

- [1]. Costa TR, Felisberto-Rodrigues C, Meir A, Prevost MS, Redzej A, Trokter M, Waksman G. S. Secretion systems in Gram-negative bacteria: structural and mechanistic insights. *Nat Rev Microbiol*. 2015 Jun;13(6):343-59.
- [2]. Cornelis, G. The type III secretion injectisome. *Nat Rev Microbiol* 4, 811–825 (2006).
- [3]. Green ER, Mecsas J. Bacterial Secretion Systems: An Overview. *Microbiol Spectr*. 2016 Feb; 4(1):10.1128/microbiolspec.VMBF-0012-2015.
- [4]. Mittal R, Peak-Chew SY, Sade RS, Vallis Y, McMahon HT. The acetyltransferase activity of the bacterial toxin YopJ of *Yersinia* is activated by eukaryotic host cell inositol hexakisphosphate. *J Biol Chem*. 2010 Jun 25;285(26):19927-34.
- [5]. Hatakeyama M. *Helicobacter pylori* CagA and gastric cancer: a paradigm for hit-and-run carcinogenesis. *Cell Host Microbe*. 2014 Mar 12;15(3):306-16.
- [6]. Sana TG, Hachani A, Bucior I, Soscia C, Garvis S, Termine E, Engel J, Filloux A, Bleves S. The second type VI secretion system of *Pseudomonas aeruginosa* strain PAO1 is regulated by quorum sensing and Fur and modulates internalization in epithelial cells. *J Biol Chem*. 2012 Aug 3;287(32):27095-105.
- [7]. Tan L, Yammani RR. Co-Immunoprecipitation-Blotting: Analysis of Protein-Protein Interactions. *Methods Mol Biol*. 2022;2413:145-154. doi: 10.1007/978-1-0716-1896-7_15.
- [8]. Richards AL, Eckhardt M, Krogan NJ. Mass spectrometry-based protein-protein interaction networks for the study of human diseases. *Mol Syst Biol*. 2021 Jan;17(1):e8792. doi: 10.15252/msb.20188792. PMID: 33434350; PMCID: PMC7803364.
- [9]. Duarte CEM, Euclides NC. Protein-Protein Interaction via Two-Hybrid Assay in Yeast. *Methods Mol Biol*. 2024;2724:193-210. doi: 10.1007/978-1-0716-3485-1_14.
- [10]. Durmuş Tekir S, Çakır T, Ardiç E, Sayılırbaş AS, Konuk G, Konuk M, Sarıyer H, Uğurlu A, Karadeniz İ, Özgür A, Sevilgen FE, Ülgen KÖ. PHISTO: pathogen-host interaction search tool. *Bioinformatics*. 2013 May 15;29(10):1357-8. doi: 10.1093/bioinformatics/btt137. Epub 2013 Mar 20.
- [11]. Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, Gable AL, Fang T, Doncheva NT, Pyysalo S, Bork P, Jensen LJ, von Mering C. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res*. 2023 Jan 6;51(D1):D638-D646. doi: 10.1093/nar/gkac1000.
- [12]. Orchard S, Ammari M, Aranda B, Breuza L, Briganti L, Broackes-Carter F, Campbell NH, Chavali G, Chen C, del-Toro N, Duesbury M, Dumousseau M, Galeota E, Hinz U, Iannuccelli M, Jagannathan S, Jimenez R, Khadake J, Lagreid A, Licata L, Lovering RC, Meldal B, Melidoni AN, Milagros M, Peluso D, Perfetto L, Porras P, Raghunath A, Ricard-Blum S, Roehrich B, Stutz A, Tognolli M, van Roey K, Cesareni G, Hermjakob H. The MIntAct project—IntAct as a common curation platform for 11 molecular interaction databases. *Nucleic Acids Res*. 2014 Jan;42(Database issue):D358-63. doi: 10.1093/nar/gkt1115. Epub 2013 Nov 13.
- [13]. Pierce BG, Wiehe K, Hwang H, Kim BH, Vreven T, Weng Z. ZDOCK server: interactive docking prediction of protein-protein complexes and symmetric multimers. *Bioinformatics*. 2014 Jun 15;30(12):1771-3. doi: 10.1093/bioinformatics/btu097. Epub 2014 Feb 14. PMID: 24532726;
- [14]. Yan Y, Zhang D, Zhou P, Li B, Huang SY. HDock: a web server for protein-protein and p

- rotein-DNA/RNA docking based on a hybrid strategy. *Nucleic Acids Res.* 2017 Jul 3;45(W1): W365-W373. doi: 10.1093/nar/gkx407.
- [15]. Abramson J, Adler J, Dunger J, Evans R, Green T, Pritzel A, Ronneberger O, Willmore L, Ballard AJ, Bambrick J, Bodenstein SW, Evans DA, Hung CC, O'Neill M, Reiman D, Tunyasuvunakool K, Wu Z, Žemgulytė A, Arvaniti E, Beattie C, Bertolli O, Bridgland A, Cherepanov A, Congreve M, Cowen-Rivers AI, Cowie A, Figurnov M, Fuchs FB, Gladman H, Jain R, Khan YA, Low CMR, Perlin K, Potapenko A, Savy P, Singh S, Stecula A, Thillaisundaram A, Tong C, Yakneen S, Zhong ED, Zielinski M, Židek A, Bapst V, Kohli P, Jaderberg M, Hassabis D, Jumper JM. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature.* 2024 Jun;630(8016):493-500. doi: 10.1038/s41586-024-07487-w. Epub 2024 May 8.
- [16]. Krissinel E, Henrick K. Inference of macromolecular assemblies from crystalline state. *J Mol Biol.* 2007 Sep 21;372(3):774-97. doi: 10.1016/j.jmb.2007.05.022. Epub 2007 May 13.
- [17]. Buchfink B, Reuter K, Drost HG. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat Methods.* 2021 Apr;18(4):366-368. doi: 10.1038/s41592-021-01101-x. Epub 2021 Apr 7.
- [18]. Kelk P, Johansson AClaesson R, Hånström L, Kalfas S2003.Caspase 1 Involvement in Human Monocyte Lysis Induced by Actinobacillus actinomycetemcomitans Leukotoxin. *Infect Immun* 71:https://doi.org/10.1128/iai.71.8.4448-4455.2003
- [19]. Kachlany SC, Schwartz AB, Balashova NV, Hioe CE, Tuen M, Le A, Kaur M, Mei Y, Rao J. Anti-leukemia activity of a bacterial toxin with natural specificity for LFA-1 on white blood cells. *Leuk Res.* 2010 Jun;34(6):777-85. doi: 10.1016/j.leukres.2009.08.022. Epub 2009 Sep 10.
- [20]. Hioe CE, Tuen M, Vasiliver-Shamis G, Alvarez Y, Prins KC, Banerjee S, et al. (2011) HIV Envelope gp120 Activates LFA-1 on CD4 T-Lymphocytes and Increases Cell Susceptibility to LFA-1-Targeting Leukotoxin (LtxA). *PLoS ONE* 6(8): e23202. <https://doi.org/10.1371/journal.pone.0023202>
- [21]. Iorenz, H.M.; Grunke, M.; Hieronymus, T.; Herrmann, M.; Kuhnel, A.; Manger, B.; Kalden, J. R. In vitro apoptosis and expression of apoptosis-related molecules in lymphocytes from patients with systemic lupus erythematosus and other autoimmune diseases. *Arthritis Rheum.* 1997, 40, 306–317. <https://doi.org/10.1002/art.1780400216>
- [22]. Hatakeyama M. Oncogenic mechanisms of the Helicobacter pylori CagA protein. *Nat Rev Cancer.* 2004 Sep;4(9):688-94.
- [23]. Jiang, L., Wang, P., Song, X. et al. Salmonella Typhimurium reprograms macrophage metabolism via T3SS effector SopE2 to promote intracellular replication and virulence. *Nat Commun* 12, 879 (2021). <https://doi.org/10.1038/s41467-021-21186-4>
- [24]. Swietnicki W. Secretory System Components as Potential Prophylactic Targets for Bacterial Pathogens. *Biomolecules.* 2021 Jun 15;11(6):892. doi: 10.3390/biom11060892.
- [25]. Navarro L, Koller A, Nordfelth R, Wolf-Watz H, Taylor S, Dixon JE. Identification of a molecular target for the Yersinia protein kinase A. *Mol Cell.* 2007 May 25;26(4):465-77.
- [26]. Sanya DRA, Onésime D, Vizzarro G, Jacquier N. Recent advances in therapeutic target identification and development of treatment strategies towards Pseudomonas aeruginosa infections. *BMC Microbiol.* 2023 Mar 30;23(1):86. doi: 10.1186/s12866-023-02832-x.
- [27]. Qin S, Xiao W, Zhou C, Pu Q, Deng X, Lan L, Liang H, Song X, Wu M. Pseudomonas aeruginosa

- 1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
- ruginosa: pathogenesis, virulence factors, antibiotic resistance, interaction with host, technology advances and emerging therapeutics. *Signal Transduct Target Ther*. 2022 Jun 25;7(1):199. doi: 10.1038/s41392-022-01056-1.
- [28]. Hicks SW, Galán JE. Exploitation of eukaryotic subcellular targeting mechanisms by bacterial effectors. *Nat Rev Microbiol*. 2013 May;11(5):316-26.
- [29]. Eckart RA, Bisle S, Schulze-Luehrmann J, Wittmann I, Jantsch J, Schmid B, Berens C, Lührmann A. Antiapoptotic activity of *Coxiella burnetii* effector protein AnkG is controlled by p32-dependent trafficking. *Infect Immun*. 2014 Jul;82(7):2763-71. doi: 10.1128/IAI.01204-13. Epub 2014 Apr 14.
- [30]. Ammari MG, Gresham CR, McCarthy FM, Nanduri B. HPIDB 2.0: a curated database for host-pathogen interactions. *Database (Oxford)*. 2016 Jul 3;2016:baw103. doi: 10.1093/database/baw103.
- [31]. Bi D, Liu L, Tai C, Deng Z, Rajakumar K, Ou HY. SecReT4: a web-based bacterial type IV secretion system resource. *Nucleic Acids Res*. 2013 Jan;41(Database issue):D660-5. doi: 10.1093/nar/gks1248. Epub 2012 Nov 28
- [32]. Chen, R., Chen, Z., Lu, Q., Cao, X., Zhong, J., Hu, Y., Zhang, J., Liu, S., Zhao, Z., Wu, J., Chen, C., & Wang, Y. (n.d.). TxSEdb: a manually annotated database for experimentally validated secreted proteins in Gram-negative bacteria. Manuscript submitted for publication.
- [33]. Rose PW, Prlić A, Altunkaya A, Bi C, Bradley AR, Christie CH, Costanzo LD, Duarte JM, Dutta S, Feng Z, Green RK, Goodsell DS, Hudson B, Kalro T, Lowe R, Peisach E, Randle C, Rose AS, Shao C, Tao YP, Valasatava Y, Voigt M, Westbrook JD, Woo J, Yang H, Young JY, Zardecki C, Berman HM, Burley SK. The RCSB protein data bank: integrative view of protein, gene and 3D structural information. *Nucleic Acids Res*. 2017 Jan 4;45(D1):D271-D281. doi: 10.1093/nar/gkw1000. Epub 2016 Oct 27.
- [34]. Varadi M, Anyango S, Deshpande M, Nair S, Natassia C, Yordanova G, Yuan D, Stroe O, Wood G, Laydon A, Židek A, Green T, Tunyasuvunakool K, Petersen S, Jumper J, Clancy E, Green R, Vora A, Lutfi M, Figurnov M, Cowie A, Hobbs N, Kohli P, Kleywegt G, Birney E, Hassabis D, Velankar S. AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Res*. 2022 Jan 7;50(D1):D439-D444. doi: 10.1093/nar/gkab1061.
- [35]. UniProt Consortium. UniProt: a worldwide hub of protein knowledge. *Nucleic Acids Res*. 2019 Jan 8;47(D1):D506-D515. doi: 10.1093/nar/gky1049.
- [36]. Sayers EW, Bolton EE, Brister JR, Canese K, Chan J, Comeau DC, Connor R, Funk K, Kelly C, Kim S, Madej T, Marchler-Bauer A, Lanczycki C, Lathrop S, Lu Z, Thibaud-Nissen F, Murphy T, Phan L, Skripchenko Y, Tse T, Wang J, Williams R, Trawick BW, Pruitt KD, Sherry ST. Database resources of the national center for biotechnology information. *Nucleic Acids Res*. 2022 Jan 7;50(D1):D20-D26. doi: 10.1093/nar/gkab1112.
- [37]. Krogh A, Larsson B, von Heijne G, Sonnhammer EL. Predicting transmembrane protein topology with a hidden Markov model: application to complete genomes. *J Mol Biol*. 2001 Jan 19;305(3):567-80. doi: 10.1006/jmbi.2000.4315.
- [38]. Nielsen H, Teufel F, Brunak S, von Heijne G. SignalP: The Evolution of a Web Server. *Methods Mol Biol*. 2024;2836:331-367. doi: 10.1007/978-1-0716-4007-4_17.
- [39]. Yu NY, Wagner JR, Laird MR, Melli G, Rey S, Lo R, Dao P, Sahinalp SC, Ester M, Foster LJ, Brinkman FS. PSORTb 3.0: improved protein subcellular localization prediction with refin

ed localization subcategories and predictive capabilities for all prokaryotes. *Bioinformatics*. 2010 Jul 1;26(13):1608-15. doi: 10.1093/bioinformatics/btq249. Epub 2010 May 13.

[40]. Franz M, Lopes CT, Fong D, Kucera M, Cheung M, Siper MC, Huck G, Dong Y, Sumer O, Bader GD. Cytoscape.js 2023 update: a graph theory library for visualization and analysis. *Bioinformatics*. 2023 Jan 1;39(1):btad031. doi: 10.1093/bioinformatics/btad031.

[41]. Deqing Li, Honghui Mei, Yi Shen, Shuang Su, Wenli Zhang, Junting Wang, Ming Zu, Wei Chen, ECharts: A declarative framework for rapid construction of web-based visualization, *Visual Informatics*, Volume 2, Issue 2, 2018, Pages 136-146, ISSN 2468-502X, <https://doi.org/10.1016/j.visinf.2018.04.011>.

[42]. Sehnal D, Bittrich S, Deshpande M, Svobodová R, Berka K, Bazgier V, Velankar S, Burley SK, Koča J, Rose AS. Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures. *Nucleic Acids Res*. 2021 Jul 2;49(W1):W431-W437. doi: 10.1093/nar/gkab314.

For Peer Review